

Round Monotonic Scheduling

```
C rms.c > main()
1 #include <stdio.h>
2 typedef struct {
3     int id, execution, period, remaining;
4 } Task;
5 int main() {
6     int n, hyperperiod;
7     printf("Enter number of tasks: ");
8     scanf("%d", &n);
9     Task t[n];
10    for (int i = 0; i < n; i++) {
11        t[i].id = i + 1;
12        printf("Task %d Execution Time: ", i + 1);
13        scanf("%d", &t[i].execution);
14        printf("Task %d Period: ", i + 1);
15        scanf("%d", &t[i].period);
16        t[i].remaining = 0;
17    }
18    printf("Enter total simulation time: ");
19    scanf("%d", &hyperperiod);
20    printf("\nRMS Scheduling:\n");
21    for (int time = 0; time < hyperperiod; time++) {
22        // Release tasks
23        for (int i = 0; i < n; i++) {
24            if (time % t[i].period == 0) {
25                t[i].remaining = t[i].execution;
26            }
27        }
28        int selected = -1;
29        for (int i = 0; i < n; i++) {
30            if (t[i].remaining > 0) {
31                if (selected == -1 || t[i].period < t[selected].period) {
32                    selected = i;
33                }
34            }
35        }
36        if (selected != -1) {
37            printf("Time %d: Task %d\n", time, t[selected].id);
38            t[selected].remaining--;
39        } else {
40            printf("Time %d: Idle\n", time);
41        }
42    }
43 }
```

```
cd "/Users/sharan/Desktop/OS LAB/" && gcc rms.c -o rms && "/Users/sharan/Desktop/OS LAB/"rms
• → OS LAB cd "/Users/sharan/Desktop/OS LAB/" && gcc rms.c -o rms && "/Users/sharan/Desktop/OS LAB/"rms
Enter number of tasks: 4
Task 1 Execution Time: 2
Task 1 Period: 3
Task 2 Execution Time: 5
Task 2 Period: 7
Task 3 Execution Time: 3
Task 3 Period: 8
Task 4 Execution Time: 2
Task 4 Period: 9
Enter total simulation time: 7

RMS Scheduling:
Time 0: Task 1
Time 1: Task 1
Time 2: Task 2
Time 3: Task 1
Time 4: Task 1
Time 5: Task 2
Time 6: Task 1
○ → OS LAB
```